Recurrent Neural Networks

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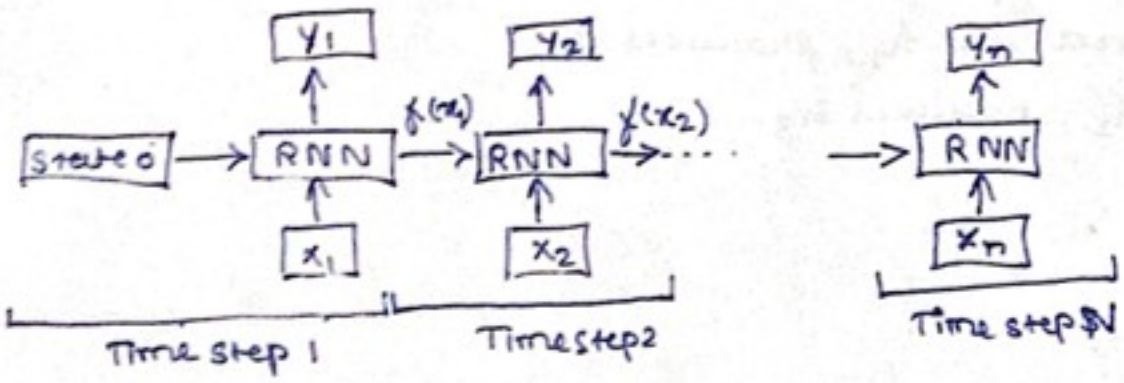

1) RNNs map one sequence of input to 1 sequence of output.
For 1st step, state 0 and input state x_1 produce output y_1

2) For step 2 :- output of step 1 i.e. hidden state 1 along with x_2 produce output y_2

3) Problem :- i) This process is slow for long sequence as previous state output is required for next state

ii) Not parallelizable

iii) Vanishing gradient :- chain rule :- $\frac{dy_n}{dx} = \frac{dy_1}{df} \cdot \frac{df}{dy_2} \cdot \frac{dy_2}{df} \cdot \frac{df}{dy_3}$

we try to minimise the gradient (slope of loss function)

Due to chain rule $\frac{dy_1}{df} \approx 0.05$, $\frac{df}{dy_2} \approx 0.4$, $\frac{dy_2}{df} \approx 0.1$, $\frac{df}{dy_3} \approx 0.05$

Due to chain of multiplication $\frac{dy_n}{dx}$ becomes smaller and smaller.

i.e. gradient (slope) becomes flat. Hence loss function is stuck at local minima (or maxima if we maximize)

iv) Difficulty in accessing information long time ago.

Due to transformations done repeatedly by $f(x_t)$ the hidden state after a long sequence has very little or no information of original input state.

So in a text of 1000s of words, relationship between words far apart is missed.

A simple recurrence is $h_t = \phi(W_{xh} x_t + W_{hh} h_{t-1} + b_h)$

$$y_t = W_{hy} h_t + b_y$$

where W_{xh} maps input x_t to hidden state

W_{hh} carries information forward

ϕ is usually tanh or ReLU

e.g. reading a sentence one word at a time while keeping notes in a small notebook. Each new word updates the notebook and notebook influences how the next word is predicted.

Unrolling an RNN

step1 reads x_1, produces h_1
step2 reads x_2, ~~produces~~ uses h_1, produces h_2
step3 reads x_3, uses h_2, produces h_3

Important in

1) speech recognition
2) Language modelling
3) time series forecasting
Because they naturally model temporal ~~differences~~ dependence

Training -

RNNs are trained with backpropagation through time (BPTT) which means we unroll the network across time steps and then apply backpropagation through ~~time~~ the whole chain